

Roman Urdu Medical Triage in Pakistan

Methodology writing points | Embedding configuration C

01 Study design and scope

Describe an offline research prototype for Roman Urdu medical triage in Pakistan, restricted to cardiac complaints and triage levels 1-4. State that the data is synthetic and the system has not been clinically validated.

02 Dataset and provenance

Use the active experiment: 2,252 synthetic cardiac records from `cardiac_english_2252.csv`. Describe complaint generation, labels, class distribution and structured fields. Training translations were prepared using `gpt-4o-mini`; they are not local Ollama translations.

03 Roman Urdu normalization

Explain rule-based spelling replacement, fuzzy clinical vocabulary matching and canonical forms. Keep this distinct from the English classifier preprocessing, which skips Roman Urdu normalization.

04 Local translation and refusal handling

At prediction time, translate complaints to English using Ollama on the local computer, with `llama3.2` preferred and temperature 0.0. Record the actual selected model. Describe refusal filtering and unsuccessful translation handling.

05 Anatomical consistency gate

Verify deterministically that named body parts in the normalized complaint are preserved in the English translation. Explain that a failed gate or failed translation prevents a triage score.

Methodology continued

06 Learned stop-word removal

Explain frequency, normalized mutual information and Cramer's V criteria, clinical-term protection and saved per-token reports. Evaluation learns the list on training rows. The active English bundle contains 68 deployed stop words.

07 Sentence embeddings and feature fusion

Name sentence-transformers/paraphrase-multilingual-MiniLM-L12-v2. It produces 384 L2-normalized dimensions. Add six standardized numeric inputs and 20 one-hot categorical columns, for 410 total classifier features.

08 Classifier and configuration

Describe configuration C: preprocessed English embeddings plus structured features, classified by Logistic Regression with balanced class weights and max_iter=1200. Use the existing English bundle and identify its 2026-09-09 training date.

09 Evaluation and ablation

Report the recorded stratified 80/20 split (1,801 training and 451 test rows), seed 42, confusion matrix, accuracy and under/over-triage. Compare raw B against preprocessed C. Recorded C scores are 80.49%, 12.64% and 6.87%; B scores are 80.71%, 13.30% and 5.99%, respectively.

10 Separate studies and limitations

Describe cluster similarity and nearest-other-complaint agreement as representation diagnostics. The 185-row e5-small baseline, two targets and stratified five-fold protocol are a separate experiment. Disclose the training/serving translation mismatch, structured preprocessing fitted before splitting, and full-data stop-word export beside a split-trained classifier.